

INVESTMENT ASSIGNMENT

SUBMISSION

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Business Understanding

Project Brief

I work for Spark Funds, an asset management company. Spark Funds wants to make investments in a few companies. The CEO of Spark Funds wants to understand the global trends in investments so that she can take the investment decisions effectively.

Business and Data Understanding

Spark Funds has two minor constraints for investments:

- It wants to invest between **5 to 15 million USD** per round of investment
- It wants to invest only in **English-speaking countries** because of the ease of communication with the companies it would invest in
- Spark Funds wants to invest where most **other investors are investing**. This pattern is often observed among early stage startup investors.

What is Spark Funds' business objective?

The objective is to identify the best sectors, countries, and a suitable investment type for making investments. The overall strategy is to invest where others are investing, implying that the 'best' sectors and countries are the ones 'where most investors are investing'.

Problem Solving Methodology :Analysis work flow chart

1) Data Cleaning

Merge the two data frames(companies and rounds data) so that all variables (columns) in the **companies** frame are added to the **rounds2** data frame.

2) Funding Type Analysis

- Calculate the **most representative value of the investment amount** for each of the four funding types (venture, angel, seed, and private equity) and which investment type is the most suitable for Spark Funds?
- Since Spark Funds wants to invest between **5 to 15 million USD** per investment round, so we find which investment type is the most suitable for it. For further analysis, filter the data so it only contains the chosen investment type.

3) Country Analysis

- For the chosen investment type, make a data frame with the top nine countries (based on the total investment amount each country has received)

4) Sector Analysis

- For sector analysis, refer to one of the **eight main sectors** listed in the mapping file. This is to simplify the analysis by grouping the numerous category lists in the mapping file.
- Extract the primary sector of each category list.
- Use the **mapping file** to map each primary sector to one of the eight main sectors (
- Create three separate data frames for each of the top three English countries containing the observations of the chosen funding type falling within the **5-15 million USD** range.

Work Flow Chart

Data Cleaning → *Funding Type Analysis* → *Country Analysis* → *Sector Analysis*

Step 1: Data Cleaning

- 1) Load the companies and rounds data into two data frames and name them **companies** and **rounds2** respectively.
- 2) Changing the encoding type because of special characters. And then solving the multiple encoding issue.
- 3) Check for **Unique companies** are present in the two files companies and rounds2.
- 4) Merge the two data frames so that all variables (columns) in the **companies** frame are added to the **rounds2** data frame. Name the merged frame **master_frame**. How many observations are present in **master_frame**?
- 5) After this, we will need to work only with the **master frame**.
- 6) Inspecting the Null values , column-wise. This is because for Sparks Funds, we are mostly driving our analysis based on funding round type, category, country etc. Hence, so many columns present in the master_frame are not needed, we will drop those columns.
- 7) Inspecting the Null values , row-wise. For the remaining columns of master_frame dataframe, we can see that there are columns which still have null counts, let's drop those rows and inspect the dataframe again.
- 8) Dropping unnecessary columns and rows

Step 2 & 3 : Funding Type Analysis & Country Analysis

Step 2: Funding Type Analysis

- 1) Spark Funds wants to choose one of these four investment types(venture, angel, seed, and private equity) for each potential investment they will make. So let's observe and see how many funding types are present in master_frame and then retain the rows with above-mentioned investment types.
- 2) Retaining the rows with only four investment types venture, seed, angel & private_equity
- 3) Calculate the average investment amount for each of the four funding types.
- 4) Converting \$ to million \$ for ease the task of comparison
- 5) Only Venture investment type fall within the between **5 to 15 million USD** per round of investment range. Retaining rows with only venture type. As Spark Funds wants to invest between 5 to 15 million USD per investment round

Step 3: Country Analysis

- 1) Spark Funds wants to see the top nine countries which have received the highest total funding (across ALL sectors for the chosen investment type)
- 2) For the chosen investment type, make a data frame with the top nine countries (based on the total investment amount each country has received)

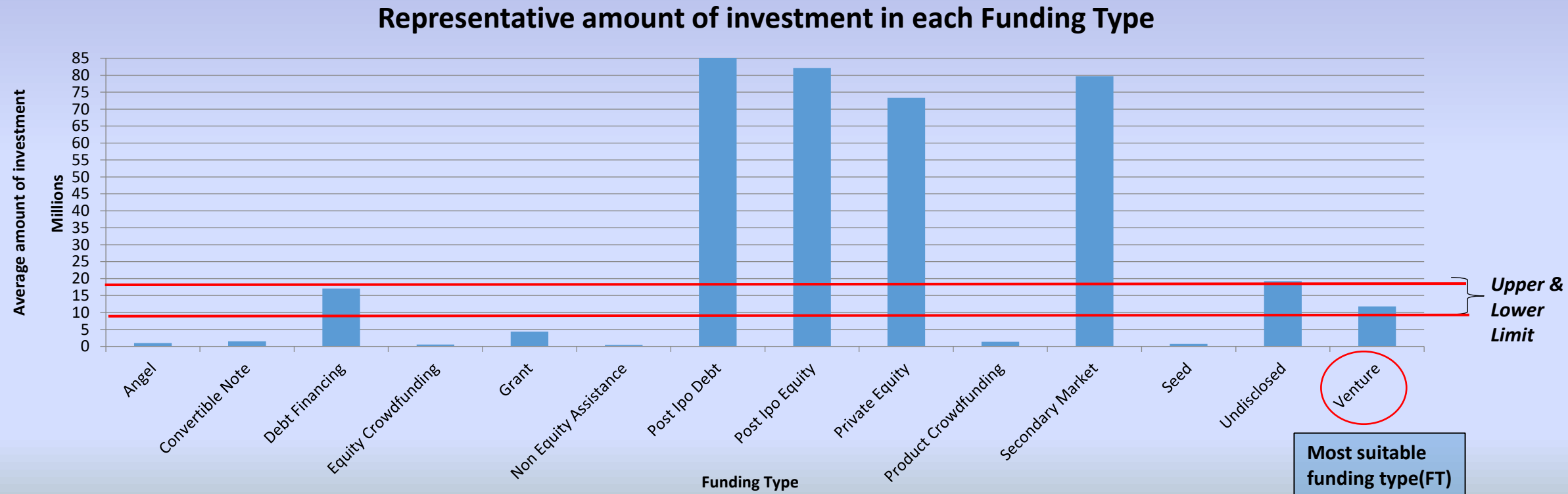
Retaining rows with only USA, GBR and IND country_codes. As SparksFunds wants to invest in only top three English speaking countries.

Step 4 : Sector Analysis

- 1) Extract the primary sector of each category
- 2) Extract the primary sector value into *category_list* column. According to the business rule the first string before the vertical bar will be considered the primary sector.
- 3) Map each primary sector to one of the eight main sector , use the mapping file 'mapping.csv' to map each primary sector to one of the eight main sectors (Note that 'Others' is also)
- 4) *Reshaping the mapping dataframe to merge with the master_frame dataframe. Using melt() function to unpivot the table.*
- 5) *Retaining the rows which have main_sector values*
- 6) Create DataFrames D1, D2, D3 based on three countries containing the observations of funding type FT falling within the 5-15 million USD range. The three data frames should contain:
 - a) All the columns of the master_frame along with the primary sector and the main sector
 - b) The total number (or count) of investments for each main sector in a separate column
 - c) The total amount invested in each main sector in a separate column
- 7) *Find the total number of investments (count) and the total amount of investment (USD)*
- 8) *Find the Top sector, second-top, third-top for each of the three dataframes (based on count of investments)*
- 9) *Number of investments in the top, second-top, third-top sector in D1*
- 10) *For the top three sector USA,GBR & IND, which company received the highest investment?*
- 11) *For the second top three sector USA,GBR & IND, which company received the highest investment?*

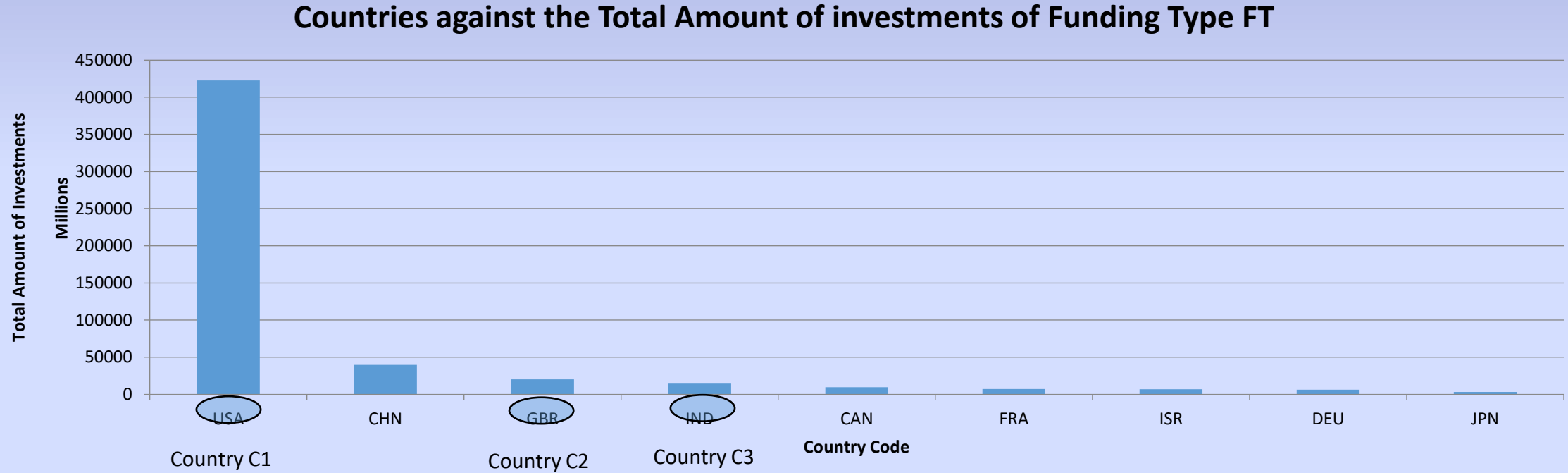
Results: Plot 1

A plot showing **the representative amount of investment** in each funding type. This chart should make it clear that a certain funding type (FT) is best suited for Spark Funds.



Results: Plot 2

A plot showing the top 9 countries against the total amount of investments of funding type FT.

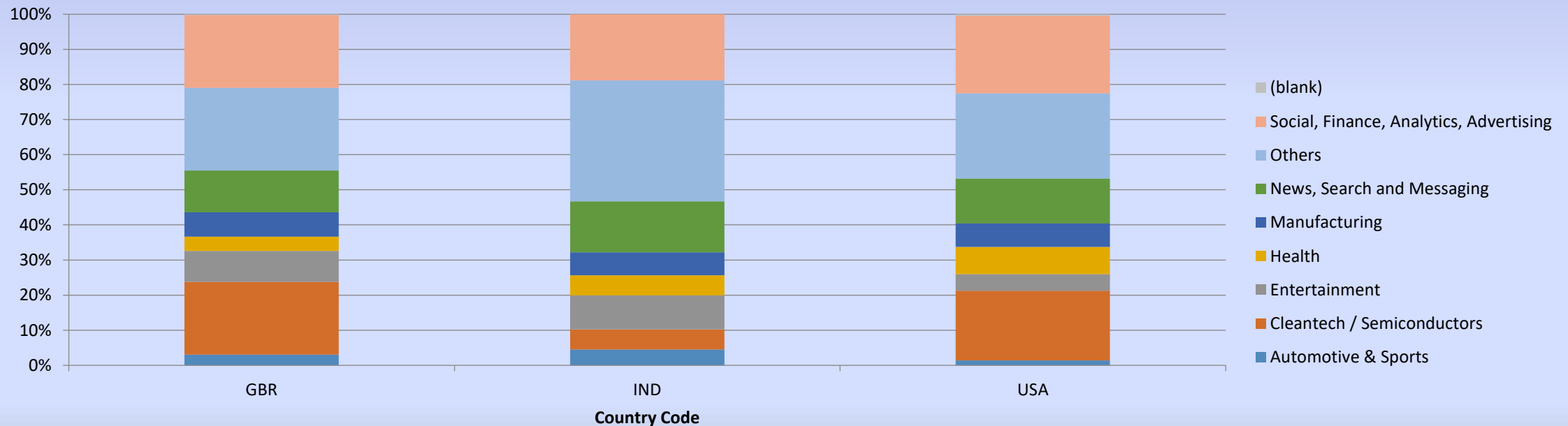


The Top 3 English speaking countries

Results: Plot 3

- A plot showing the number of investments in the top 3 sectors of the top 3 countries on one chart (for the chosen investment type FT).

Number of Investments in the Top 3 sectors of the Top 3 countries



Conclusions

- Based on the data analysis performed, SparksFunds should invest in -
- Funding type - Venture.
- Countries - USA, Britain and India, respectively.
- Top two sectors to invest in are - Others and Social, Finance, Analytics, Advertising.